Hints: Limits – 2. Recurrences solvable in closed form

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. By AM–GM, $x_n = \frac{1}{2}(x_{n-1} + \frac{c}{x_{n-1}}) \geqslant \sqrt{c}$ for every $n \geqslant 1$, whatever $x_0 > 0$ is. Use this to show that from $n = 1$ on the sequence is decreasing, hence convergent; the limit $L > 0$ satisfies $L = \frac{1}{2}(L + c/L)$.

Hint 2. For the speed, compute $x_n - \sqrt{c} = \dfrac{(x_{n-1} - \sqrt{c})^2}{2x_{n-1}} \leqslant \dfrac{(x_{n-1} - \sqrt{c})^2}{2\sqrt{c}}$ (for $n \geqslant 2$). So the error is squared at each step (quadratic convergence): with $e_n = (x_n - \sqrt{c})/(2\sqrt{c})$ one has $e_n \leqslant e_{n-1}^2$, hence $e_n \leqslant e_1^{2^{n-1}}$, and the number of correct digits roughly doubles each step.

Hint 3. Newton's method for $g(x) = x^2 - c$ is $x_n = x_{n-1} - g(x_{n-1})/g'(x_{n-1})$; check that this is exactly the given recurrence.

Problem 2

Hint 1. Following the suggestion, express $a_n = (b_n^2 - 1)/24$ and compute $b_{n+1}^2 = 1 + 24a_{n+1}$ in terms of b_n. It turns out to be a perfect square.

Hint 2. One gets $b_{n+1}^2 = \left(\frac{b_n+3}{2}\right)^2$, so $b_{n+1} = \frac{b_n+3}{2}$ with $b_1 = 5$, a linear recurrence: $b_n - 3 = 2^{2-n}$.

Hint 3. Hence $a_n = \dfrac{(3 + 2^{2-n})^2 - 1}{24} = \dfrac{1}{3} + \dfrac{1}{2^n} + \dfrac{2}{3 \cdot 4^n} = \dfrac{(2^n + 1)(2^n + 2)}{3 \cdot 4^n}$.

Problem 3

Note. The recurrence for b must be used for $n \geqslant 0$ (otherwise b_1 is undefined); so $b_1 = a_0 - b_0 = 1$. The recurrence for a is for $n \geqslant 0$ as well.

Hint 1. Compute a few terms: $a_n = 1, 2, 5, 13, 34, \ldots$ and $b_n = 0, 1, 1, 4, 9, 25, \ldots$. Recognize Fibonacci numbers: conjecture $a_n = F_{2n+1}$ and $b_n = F_n^2$.

Hint 2. Get rid of the root: $(2a_{n+1} - 3a_n)^2 = 5a_n^2 - 4$ gives $a_{n+1}^2 - 3a_n a_{n+1} + a_n^2 + 1 = 0$. This relation is symmetric in a_n, a_{n+1}, so both a_{n+1} and a_{n-1} are roots of $t^2 - 3a_n t + a_n^2 + 1 = 0$; a_{n+1} is the larger root and $a_{n-1} < a_n \leqslant a_{n+1}$, so by Vieta $a_{n+1} = 3a_n - a_{n-1}$.

Hint 3. $F_{2n+3} = 3F_{2n+1} - F_{2n-1}$, so $a_n = F_{2n+1}$. Then prove $b_n = F_n^2$ by induction using the identity $F_n^2 + F_{n+1}^2 = F_{2n+1}$, i.e. $b_{n+1} = a_n - b_n$. Finally $b_{2n+1} = F_{2n+1}^2 = a_n^2$.

Problem 4

Hint 1. Theorem 1: the characteristic roots of $\lambda^2 - 6\lambda + 8$ are 2 and 4; the right-hand side $n = n \cdot 1^n$ and 1 is not a root, so look for a particular solution $An + B$.

Hint 2. Substituting gives $A = \frac{1}{3}$, $B = \frac{10}{9}$. Then fit $a_n = C2^n + D4^n + \frac{n}{3} + \frac{10}{9}$ to $a_0 = 1$, $a_1 = 2$.

Hint 3. $C = -\frac{1}{2}$, $D = \frac{7}{18}$, i.e. $a_n = \dfrac{7 \cdot 4^n - 9 \cdot 2^n + 6n + 20}{18}$ (check: $a_2 = 6$).

Problem 5

Hint 1. Write $\alpha = \cos\theta$ with $\theta \in [0, \pi]$ and show by induction that $a_n = \cos n\theta$, using $\cos(n + 2)\theta + \cos n\theta = 2\cos\theta\cos(n + 1)\theta$. Then $|a_n| \leqslant 1$ is immediate.